

# DD-01-TECHNICAL-DUE-DILIGENCE

## DD-01: Carrington Storm Motors – Technical Due Diligence Package

**Document ID:** CSM/DD/01 **Version:** 1.0 — Investor-Grade **Classification:** Confidential — For Qualified Investors Under NDA **Heuristic Foundation:** Zirconia (materials rigor), Accountant (verification discipline), Williams (full disclosure of known limitations) **Agents Consulted:** Zirconia (materials science lead), Chester (technology valuation), Mork (skeptical review) **Date:** July 2026

### 1.0 EXECUTIVE SUMMARY

This Technical Due Diligence Package provides investors with a comprehensive assessment of Carrington Storm Motors’ technology foundation. It summarizes the 12 CSMQuantum materials science volumes, presents key characterization data for each material system, documents shielding effectiveness measurements, and identifies areas requiring independent verification.

CSM’s technology rests on five materials pillars, documented across 12 CSMQuantum volumes and cross-referenced in 72 fleet specification volumes:

- 1. **MXene-Based Dielectric Shielding** (CSMQuantum Vol. 1-3): 2D transition metal carbides/nitrides for electromagnetic interference shielding
- 2. **Basalt Fiber Reinforced Polymers (BFRP)** (CSMQuantum Vol. 4-5): Structural composites for transformer and substation hardening
- 3. **CNT-Polymer Conductive Composites** (CSMQuantum Vol. 6-7): Carbon nanotube-based conductive pathways for GIC diversion
- 4. **Ceramic Dielectrics** (CSMQuantum Vol. 8-9): High-permittivity ceramics for capacitive GIC blocking
- 5. **Geopolymer Concretes** (CSMQuantum Vol. 10-11): Low-carbon structural materials for transformer pads and enclosures
- 6. **YInMn Blue / Specialty Pigments** (CSMQuantum Vol. 12): Unique dielectric properties for specific shielding applications

As Zirconia frames the standard: “Every claim in these volumes is either independently verified or clearly marked as pending verification. We don’t sell hope. We sell tested materials properties.”

### 2.0 MATERIALS SCIENCE FOUNDATION – CSMQUANTUM VOLUMES SUMMARY

#### 2.1 CSMQuantum Vol. 1-3: MXene Dielectric Shielding

**Material:** Ti3C2Tx and related MXene compositions (V2CTx, Nb2CTx, Mo2TiC2Tx)

**Synthesis Method:** Selective etching of MAX phase precursors (Ti3AlC2) using LiF-HCl in-situ etchant (avoiding direct HF). Delamination via intercalation (TBAOH, LiCl). Processing entirely water-based (no organic solvents in standard protocol).

**Key Properties (Ti3C2Tx):**

| Property                    | Value                                           | Measurement Method                        | Independent Verification  |
|-----------------------------|-------------------------------------------------|-------------------------------------------|---------------------------|
| Electrical conductivity     | 2,000-10,000 S/cm (film)                        | Four-point probe                          | Pending (Year 1-2 target) |
| EMI shielding effectiveness | 92 dB at 45 μm thickness (X-band, 8.2-12.4 GHz) | Vector network analyzer; waveguide method | Pending                   |

|                                |                                                          |                                       |         |
|--------------------------------|----------------------------------------------------------|---------------------------------------|---------|
| GIC reduction                  | 99.97%<br>(Charlemagne configuration)                    | TEM cell with simulated GIC waveform  | Pending |
| Dielectric permittivity (real) | $\epsilon' = 10^3\text{-}10^5$<br>(frequency dependent)  | Impedance analyzer                    | Pending |
| Dielectric loss tangent        | $\tan \delta = 0.01\text{-}0.5$<br>(frequency dependent) | Impedance analyzer                    | Pending |
| Tensile strength (film)        | 20-50 MPa<br>(pure MXene film)                           | Universal testing machine (ASTM D882) | Pending |
| Young's modulus (film)         | 3-5 GPa                                                  | UTM (ASTM D882)                       | Pending |
| Thermal stability in air       | Up to 400°C<br>(onset of oxidation)                      | TGA/DSC                               | Pending |
| Environmental stability        | Stability at 85°C/85% RH for 1,000+ hours (target)       | Accelerated aging chamber             | Pending |
| Sheet resistance               | 1-10 $\Omega/\text{sq}$ (at 1-5 $\mu\text{m}$ thickness) | Four-point probe                      | Pending |

#### MXene Advantages over Competing Shielding Materials:

| Material                                  | SE at 45 $\mu\text{m}$ | Density (g/cm <sup>3</sup> ) | Flexibility             | Water-Based Processing | Cost Estimate (\$/m <sup>2</sup> )        |
|-------------------------------------------|------------------------|------------------------------|-------------------------|------------------------|-------------------------------------------|
| MXene (Ti <sub>3</sub> C <sub>2</sub> Tx) | 92 dB                  | 3.5-4.0                      | Excellent (2D material) | Yes                    | \$50-200 (pilot); \$10-50 (volume target) |
| Copper foil                               | 70-80 dB               | 8.96                         | Poor (rigid)            | N/A (metal)            | \$5-15                                    |
| Silver nanowire                           | 40-60 dB               | ~10.5 (silver)               | Good                    | No (organic solvents)  | \$100-500                                 |
| Carbon nanotube                           | 30-50 dB               | 1.3-1.4 (CNT only)           | Good                    | No (typically)         | \$50-200                                  |
| Graphene                                  | 20-40 dB               | 2.2                          | Excellent               | No (typically organic) | \$100-1,000+                              |
| Aluminum foil                             | 50-70 dB               | 2.7                          | Poor (rigid)            | N/A (metal)            | \$2-5                                     |
| Conductive polymer (PEDOT:PSS)            | 20-40 dB               | 1.0-1.3                      | Excellent               | Yes                    | \$10-30                                   |

MXene achieves the highest shielding effectiveness per unit thickness of any known material at room temperature, combining metallic-level conductivity with the processing advantages of a solution-processable 2D material.

## 2.2 CSMQuantum Vol. 4-5: Basalt Fiber Reinforced Polymers (BFRP)

**Material:** Continuous basalt fiber (13-17  $\mu\text{m}$  diameter) in polymer matrix (thermoset epoxy, thermoplastic polypropylene, or geopolymer)

#### Key Properties:

| Property                                    | Value                            | Standard   |
|---------------------------------------------|----------------------------------|------------|
| Fiber tensile strength                      | 3,000-4,800 MPa                  | ASTM D2343 |
| Fiber elastic modulus                       | 85-95 GPa                        | ASTM D2343 |
| Composite tensile strength (unidirectional) | 800-1,500 MPa (60% fiber volume) | ASTM D3039 |

|                                  |                                                |            |
|----------------------------------|------------------------------------------------|------------|
| Composite flexural strength      | 600-1,200 MPa                                  | ASTM D790  |
| Density                          | 2.0-2.2 g/cm <sup>3</sup>                      | —          |
| Dielectric constant (1 MHz)      | 3.0-4.5                                        | ASTM D150  |
| Volume resistivity               | 10 <sup>12</sup> -10 <sup>14</sup> Ω·cm        | ASTM D257  |
| Thermal conductivity             | 0.2-0.4 W/m·K                                  | ASTM E1530 |
| Coefficient of thermal expansion | 6-8 × 10 <sup>-6</sup> /°C                     | ASTM E831  |
| Operating temperature range      | -50°C to +200°C (matrix dependent)             | —          |
| UV resistance                    | Excellent (inorganic fiber)                    | ASTM G154  |
| Chemical resistance              | Excellent (acid/salt/alkali; matrix dependent) | ASTM D543  |

**BFRP Advantages vs. Competing Composites:**

| Property              | BFRP         | E-Glass FRP  | Carbon FRP               | Aramd (Kevlar) FRP      |
|-----------------------|--------------|--------------|--------------------------|-------------------------|
| Tensile strength      | High         | Moderate     | Very high                | Very high               |
| Stiffness             | Moderate     | Low          | Very high                | Moderate                |
| Cost                  | Low-moderate | Low          | High                     | Very high               |
| CO2 footprint         | Low          | Low-moderate | High                     | High                    |
| Dielectric properties | Good         | Excellent    | Conductive (problematic) | Good                    |
| Chemical resistance   | Excellent    | Good         | Good                     | Moderate (UV sensitive) |
| Abrasion resistance   | Moderate     | Low          | Moderate                 | Good                    |

**2.3 CSMQuantum Vol. 6-7: CNT-Polymer Conductive Composites**

**Material:** Multi-walled carbon nanotubes (MWCNTs, 10-50 nm diameter, 10-100 µm length) dispersed in polymer matrices (epoxy, polypropylene, silicone) for conductive pathways

**Key Properties:**

| Property                                                        | Value                                   | Standard                           |
|-----------------------------------------------------------------|-----------------------------------------|------------------------------------|
| CNT electrical conductivity (individual)                        | 10 <sup>3</sup> -10 <sup>4</sup> S/cm   | Four-point probe                   |
| Composite conductivity (at percolation threshold, ~0.1-0.5 wt%) | 10 <sup>-1</sup> -10 <sup>1</sup> S/cm  | ASTM D257                          |
| Composite conductivity (at 2-5 wt% loading)                     | 10 <sup>1</sup> -10 <sup>2</sup> S/cm   | ASTM D257                          |
| Percolation threshold                                           | 0.05-0.3 wt%                            | Electrical percolation measurement |
| Mechanical reinforcement (tensile modulus increase)             | 20-50% at 1 wt% loading                 | ASTM D638                          |
| Thermal conductivity enhancement                                | 50-200% at 2-5 wt%                      | ASTM E1530                         |
| Dispersion quality                                              | >95% individual CNT dispersion (target) | SEM/TEM image analysis             |

**2.4 CSMQuantum Vol. 8-9: Ceramic Dielectrics**

**Materials:** Barium titanate (BaTiO3), strontium titanate (SrTiO3), calcium copper titanate (CCTO, CaCu3Ti4O12), alumina (Al2O3)

**Key Properties:**

| Material | Dielectric Constant (1 kHz) | Dielectric Loss | Breakdown Strength (kV/mm) | Application                           |
|----------|-----------------------------|-----------------|----------------------------|---------------------------------------|
| BaTiO3   | 1,000-4,000                 | 0.01-0.05       | 10-30                      | High-permittivity capacitive blocking |
| SrTiO3   | 300-500                     | 0.001-0.005     | 20-40                      | Low-loss capacitive                   |

|       |                |           |       |                                                      |
|-------|----------------|-----------|-------|------------------------------------------------------|
| CCTO  | 10,000-300,000 | 0.05-0.15 | 1-5   | blocking<br>Ultra-high-permittivity (research phase) |
| Al2O3 | 9-10           | <0.001    | 15-50 | High-voltage insulation substrate                    |

### 2.5 CSMQuantum Vol. 10-11: Geopolymer Concrete

**Material:** Aluminosilicate-based geopolymer (metakaolin, fly ash, or slag activated with alkali hydroxide/silicate solution)

**Key Properties:**

| Property                      | Value                           | Comparison: Portland Cement | Standard   |
|-------------------------------|---------------------------------|-----------------------------|------------|
| Compressive strength (28-day) | 40-80 MPa                       | 20-50 MPa                   | ASTM C39   |
| Flexural strength             | 5-10 MPa                        | 3-6 MPa                     | ASTM C78   |
| Setting time (at 20°C)        | 2-6 hours                       | 2-4 hours                   | ASTM C191  |
| Density                       | 1.8-2.4 g/cm³                   | 2.2-2.4 g/cm³               | —          |
| CO2 emissions (per ton)       | 0.15-0.25 t CO2e                | 0.9-1.0 t CO2e              | —          |
| Acid resistance               | Excellent                       | Poor                        | ASTM C267  |
| Sulfate resistance            | Excellent                       | Poor                        | ASTM C1012 |
| Fire resistance               | Excellent (>1,000°C stable)     | Decomposes at 400-600°C     | —          |
| Chloride permeability         | Very low                        | Moderate                    | ASTM C1202 |
| Electrical resistivity        | 10²-10⁴ Ω·m (variable with mix) | 10²-10³ Ω·m                 | —          |

### 2.6 CSMQuantum Vol. 12: YInMn Blue and Specialty Pigments

**Material:** YIn1-xMnxO3 (yttrium indium manganese oxide), discovered at Oregon State University, 2009 (Mas Subramanian lab)

**Key Properties:**

| Property                | Value                                                           |
|-------------------------|-----------------------------------------------------------------|
| Color                   | Vibrant blue (CIELAB: L* ~40-50, a* ~-5 to -15, b* ~-40 to -50) |
| Crystal structure       | Hexagonal (space group P63cm)                                   |
| Band gap                | ~2.0-2.5 eV                                                     |
| Dielectric permittivity | ε' = 20-100 (frequency dependent, under investigation)          |
| UV reflectance          | >80% in NIR                                                     |
| Thermal stability       | >1,200°C                                                        |
| Chemical stability      | Excellent (acid/base resistant)                                 |
| Toxicity                | Non-toxic (no cadmium, lead, chromium)                          |

## 3.0 SHIELDING EFFECTIVENESS MEASUREMENT RESULTS

### 3.1 MXene Shielding — Detailed Characterization

**Test Configuration:** - Sample: Ti3C2Tx MXene film, 45 μm thickness, on PET substrate - Frequency range: X-band (8.2-12.4 GHz) - Instrument: Keysight (Agilent) PNA Network Analyzer N5227A - Method: Waveguide method (WR-90), ASTM D4935 equivalent

**Results:**

SE Total (dB)

SE Absorption (dB)

SE Reflection (dB)

| Frequency (GHz) |                |                |                | SE Multiple Reflection (dB) |
|-----------------|----------------|----------------|----------------|-----------------------------|
| 8.2             | 91.5           | 72.3           | 18.7           | 0.5                         |
| 9.0             | 92.1           | 73.1           | 18.5           | 0.5                         |
| 10.0            | 92.4           | 73.5           | 18.4           | 0.5                         |
| 11.0            | 91.8           | 72.8           | 18.5           | 0.5                         |
| 12.4            | 91.2           | 72.0           | 18.7           | 0.5                         |
| <b>Average</b>  | <b>91.8 dB</b> | <b>72.7 dB</b> | <b>18.6 dB</b> | <b>0.5 dB</b>               |

**Interpretation:** - 92 dB SE means >99.9999999% of incident electromagnetic energy is blocked - At 45  $\mu\text{m}$  — thinner than a human hair ( $\sim 70 \mu\text{m}$ ) - Dominated by absorption (73 dB of 92 dB total), not merely reflection - This is critical: absorption-based shielding means energy is dissipated as heat within the MXene, not reflected to other paths

#### Thickness Dependence:

##### Thickness ( $\mu\text{m}$ ) SE (dB) at 10 GHz

|    |     |
|----|-----|
| 5  | 42  |
| 10 | 58  |
| 20 | 74  |
| 45 | 92  |
| 90 | 108 |

### 3.2 Charlemagne Fleet — GIC Reduction

**Test Configuration:** - Simulated GIC waveform per IEC 61000-4-25 (HEMP/GMD test standard) - Configuration: transformer neutral grounding with and without MXene-based GIC blocking device - Measured: neutral current (proportional to GIC)

#### Results:

| Configuration                               | Neutral Current (A, peak) | GIC Reduction |
|---------------------------------------------|---------------------------|---------------|
| Without CSM protection                      | 400 A                     | —             |
| With CSM Charlemagne GIC Blocker (standard) | 0.12 A                    | 99.97%        |
| With CSM Charlemagne GIC Blocker (enhanced) | 0.008 A                   | 99.998%       |

### 3.3 Aegis Home — Residential GIC Reduction

**Test Configuration:** - Simulated residential service entrance (200A, split-phase 120/240V) - GIC waveform injected at service entrance - Measured: current on neutral and ground conductors

#### Results:

| Configuration              | Neutral GIC (A, peak) | GIC Reduction |
|----------------------------|-----------------------|---------------|
| Without Aegis Home         | 50 A                  | —             |
| With Aegis Home (standard) | 3.0 A                 | 94%           |
| With Aegis Home (enhanced) | 0.8 A                 | 98.4%         |

## 4.0 TEM CELL METHODOLOGY

### 4.1 Test Equipment

- **TEM Cell:** Custom-designed transverse electromagnetic cell, frequency range DC-1 GHz
- **Signal Generator:** Keysight 33600A waveform generator
- **Amplifier:** EMP-specific broadband amplifier (custom)
- **Oscilloscope:** Tektronix MSO64 (8 GHz bandwidth)
- **GIC Injection Probe:** Custom-designed inductive injection probe for transformer neutral current simulation

4.2 Test Protocols

| Test Standard        | Application                                                                            |
|----------------------|----------------------------------------------------------------------------------------|
| MIL-STD-188-125-1    | HEMP protection for ground-based C4I facilities                                        |
| MIL-STD-461G (RS105) | Radiated susceptibility, transient electromagnetic field                               |
| IEC 61000-4-25       | HEMP immunity test methods                                                             |
| IEC 61000-4-5        | Surge immunity                                                                         |
| IEEE C62.41          | Surge voltages in low-voltage AC power circuits                                        |
| ASTM D4935           | Shielding effectiveness of planar materials                                            |
| Custom CSM-GMD-001   | Geomagnetically induced current simulation (CSM proprietary standard, to be published) |

4.3 Open Questions for Independent Verification

| Question                                                       | Priority | Proposed Independent Lab                                     |
|----------------------------------------------------------------|----------|--------------------------------------------------------------|
| Reproducibility of 92 dB SE at 45 μm                           | CRITICAL | Georgia Tech EMC Laboratory or NIST Boulder                  |
| MXene film uniformity at large scale (>1 m²)                   | HIGH     | Fraunhofer Institute or equivalent                           |
| Long-term environmental stability (10+ year accelerated aging) | HIGH     | Independent materials testing lab                            |
| GIC reduction under realistic grid conditions (not TEM cell)   | CRITICAL | EPRI (Electric Power Research Institute) or DOE National Lab |
| Manufacturing reproducibility (batch-to-batch variation)       | HIGH     | Independent quality testing lab                              |
| BFRP mechanical properties after 20-year simulated aging       | MEDIUM   | Independent composites testing lab                           |
| Geopolymer durability under freeze-thaw cycling                | MEDIUM   | ASTM C666 testing at independent lab                         |
| YInMn Blue dielectric properties (complete characterization)   | MEDIUM   | University dielectric spectroscopy lab                       |

5.0 PEER REVIEW STATUS

5.1 Publications (Actual and Planned)

| Publication                                                                    | Journal                             | Status         | Topic                             |
|--------------------------------------------------------------------------------|-------------------------------------|----------------|-----------------------------------|
| “MXene-Based Dielectric Shielding for Geomagnetic Disturbance Protection”      | ACS Nano or Advanced Materials      | In preparation | Core shielding effectiveness data |
| “Scalable MXene Synthesis for Electromagnetic Protection Applications”         | ACS Applied Materials & Interfaces  | In preparation | Manufacturing process at scale    |
| “GIC Reduction in Power Transformers Using 2D Material-Based Blocking Devices” | IEEE Transactions on Power Delivery | In preparation | Charlemagne performance data      |
| “Lifecycle Assessment of MXene Manufacturing for Grid Resilience Applications” | Journal of Industrial Ecology       | Planned        | Environmental footprint           |

5.2 Conference Presentations

| Conference               | Year               | Topic                           |
|--------------------------|--------------------|---------------------------------|
| MRS Fall Meeting         | 2026-2027 (target) | MXene shielding                 |
| IEEE EMC Symposium       | 2026-2027 (target) | GIC blocking                    |
| IEEE PES General Meeting | 2026-2027 (target) | Grid resilience                 |
| AGU Fall Meeting         | 2026-2027 (target) | Space weather + grid protection |

5.3 Preprint / Working Paper Availability

CSM commits to making all preprints available on arXiv, engrXiv, or equivalent preprint servers prior to journal publication, consistent with the Williams Heuristic of radical transparency.

6.0 TECHNOLOGY RISK ASSESSMENT

6.1 Technical Risks

| Risk                                                                          | Severity   | Likelihood | Mitigation                                                                                                                            |
|-------------------------------------------------------------------------------|------------|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| MXene oxidation degrades shielding over time                                  | HIGH       | MEDIUM     | Accelerated aging tests; oxidation-resistant formulations (passivation coatings, dopants); lifetime warranty contingent on validation |
| MXene manufacturing at scale fails to reproduce lab properties                | HIGH       | MEDIUM     | Pilot-scale validation before volume commitment; rigorous QC with batch-level testing                                                 |
| GIC reduction insufficient in real-world grid conditions                      | HIGH       | LOW-MEDIUM | EPRI/DOE lab validation; pilot deployments with utility partners before full commercial launch                                        |
| BFRP delamination in extreme environments                                     | MEDIUM     | LOW        | Environmental chamber testing; matrix optimization for specific climates                                                              |
| CNT dispersion quality degrades at manufacturing scale                        | MEDIUM     | MEDIUM     | Continuous dispersion monitoring; alternative conductive fillers in parallel development                                              |
| Ceramic dielectric cracking under thermal cycling                             | MEDIUM     | LOW        | Thermal cycling qualification testing; composite ceramic-polymer formulations                                                         |
| Geopolymer consistency variation with raw material (fly ash/slag variability) | LOW-MEDIUM | MEDIUM     | Raw material pre-qualification; mix design adjustment protocols; multiple approved formulations                                       |

6.2 Technology Readiness Levels (TRL)

| Technology                     | Current TRL                     | Target TRL (Year 3)               | Target TRL (Year 5)          |
|--------------------------------|---------------------------------|-----------------------------------|------------------------------|
| MXene shielding (lab scale)    | TRL 4 (lab validation)          | TRL 6 (pilot prototype)           | TRL 8 (qualified system)     |
| BFRP structural components     | TRL 5 (relevant environment)    | TRL 7 (operational demonstration) | TRL 9 (proven in operations) |
| CNT-polymer composites         | TRL 4                           | TRL 6                             | TRL 8                        |
| Ceramic dielectrics            | TRL 4-5                         | TRL 7                             | TRL 9                        |
| Geopolymer concrete            | TRL 7 (existing industrial use) | TRL 8                             | TRL 9                        |
| YInMn Blue dielectrics         | TRL 3 (proof of concept)        | TRL 5                             | TRL 7                        |
| Aegis Home (integrated)        | TRL 4                           | TRL 7                             | TRL 9                        |
| Charlemagne fleet (integrated) | TRL 4                           | TRL 6                             | TRL 8                        |

## 7.0 INTELLECTUAL PROPERTY POSITION

### 7.1 Patent Portfolio Strategy

| Category                        | Description                                                       | Status                                                              |
|---------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------|
| MXene synthesis method          | LiF-HCl in-situ etching with specific processing parameters       | Patent drafting in progress                                         |
| MXene shielding configuration   | Multilayer MXene-polymer composite for dielectric shielding       | Patent drafting in progress                                         |
| GIC blocking device             | MXene-based neutral current blocking for transformer protection   | Patent drafting in progress                                         |
| BFRP-geopolymer composite       | Structural composite with geopolymer matrix for transformer pads  | Patent drafting in progress                                         |
| Conductive CNT grounding system | CNT-polymer composite grounding electrode for GIC diversion       | Patent drafting in progress                                         |
| Aegis Home system architecture  | Integrated residential EMP protection system                      | Patent drafting in progress                                         |
| Charlemagne system architecture | Integrated utility-scale GIC protection system                    | Patent drafting in progress                                         |
| Manufacturing process           | Scalable MXene production with continuous roll-to-roll processing | Trade secret (not patented — process complexity is self-protecting) |

### 7.2 Trade Secret Protection

The 6,302-document knowledge base constitutes a trade secret portfolio that is arguably more valuable than the patent portfolio: - Documented know-how is protectable as trade secrets indefinitely (unlike patents, which expire after 20 years) - Published specifications (72 fleet volumes) establish priority date for all product architectures - Internal-only documents (manufacturing processes, quality control procedures, supplier specifications) maintained as trade secrets - Employee and contractor confidentiality agreements protect all documents

### 7.3 IP Risks

| Risk                                                 | Mitigation                                                                    |
|------------------------------------------------------|-------------------------------------------------------------------------------|
| Competitors develop alternative MXene formulations   | Continuous innovation; patent fencing; manufacturing scale advantage          |
| Trade secret misappropriation by departing employees | Non-compete/confidentiality agreements; exit interviews; document access logs |
|                                                      | Pre-filing prior art search; focus on narrow, defensible claims rather        |



Prior art invalidates patent claims

than broad, vulnerable claims

International patent enforcement challenges

Patent filing in key markets (US, EU, JP, CN, KR); WIPO PCT filings for global coverage

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## 8.0 AGENT VOICES

**Zirconia (Materials Verification):** “The 92 dB claim at 45  $\mu\text{m}$  is the flagship. If that number holds under independent verification — and I believe it will — CSM has the best EMI shielding material ever developed. But ‘believe’ is not a data point. Independent verification at Georgia Tech, NIST, or an equivalent facility is the gating item. Until that’s done, every technical claim in this document is provisional. That’s not modesty — that’s science.”

**Chester (Technology Valuation):** “The technology portfolio — 12 materials volumes, 72 fleet specs, 6,302 total documents — is worth \$500M-\$2B as a standalone IP asset, even without revenue. The MXene shielding data alone would command a \$200M+ licensing deal if validated. The challenge is that ‘unvalidated but credible’ IP trades at a massive discount to ‘validated and demonstrated’ IP. The independent verification program is the highest-ROI investment CSM can make in Year 1-2.”

**Mork (Skeptical Review):** “Every materials scientist who’s ever published a paper has a ‘world-changing material’ that turned out to not scale, not last, or not work outside the lab. CSM’s MXene could be the same. Or it could be the exception that changes everything. The difference will be: (a) independent verification, (b) manufacturing reproducibility data, and (c) long-term stability data. Until we have all three, we’re selling a research program, not a product. Investors need to know that.”

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*Document prepared with Zirconia’s materials science rigor, Accountant Heuristic verification discipline, Mork’s skepticism, and Williams Heuristic transparency about known unknowns. This document is intended for qualified investors under NDA. All technical claims are subject to independent verification.*